

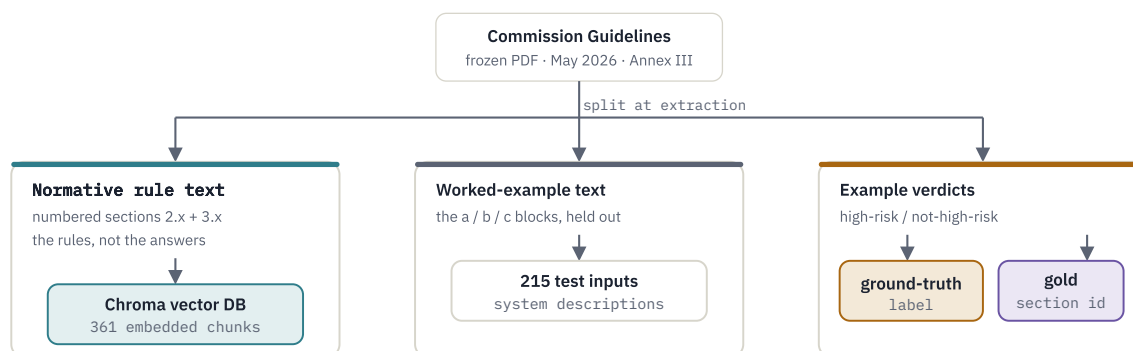
CASE STUDY 2 · PIPELINE MECHANICS

Three truth objects, and which one is in the vector database

The confusion is fair: the corpus, the correctness answer, and the retrieval answer all come from the same guidelines document. Only one of them is stored and retrieved.

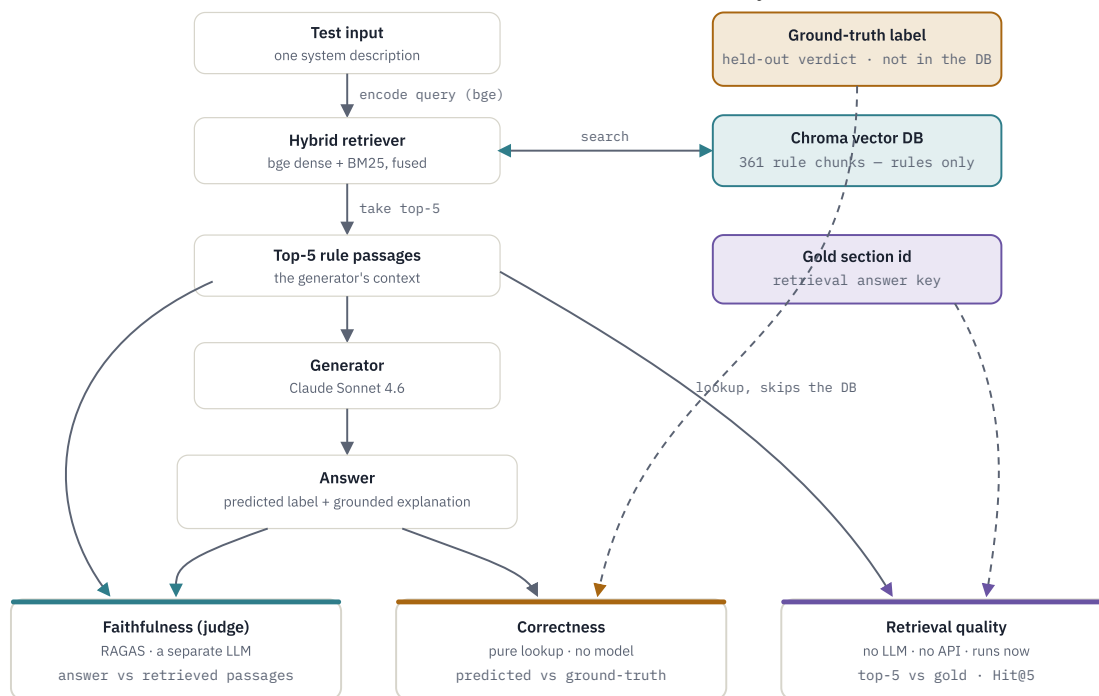
The one-line answer. The vector database holds *only* the normative rule chunks. The **ground-truth label** (is this system high-risk?) is never embedded and never retrieved. It sits in a table next to each test input and is applied once, at the very end, as a plain lookup. Retrieval and correctness are two separate lanes that only meet when the scores are compared.

Build time — where the three objects come from



One source, split once. The **rules** go into the vector database. The **test inputs**, their **correct labels**, and the **gold section pointer** come out the other side and never enter the database. The label and the gold pointer are answer keys, kept aside for scoring.

Run time — how one test input actually flows



Read the spine top to bottom: only the **retriever** ever touches the vector database. The generator sees passages, never the database. Then three scores run off to the side.

Correctness pulls the ground-truth label straight down the dashed line, bypassing the whole pipeline. **Retrieval quality** (what we are building now) needs the retriever and the gold pointer only, so it runs with **no generator and no API key**.

The three objects, side by side

RETRIEVED

Corpus chunks

The 361 normative rule paragraphs. The only thing embedded and searched. This is what the retriever fetches and the generator reads.

In the vector DB? **Yes**

LOOKED UP

Ground-truth label

The correct high-risk / not verdict for each test input, taken from the held-out worked example. Used once, to score correctness.

In the vector DB? **No**

ANSWER KEY

Gold passage

The section that governs each input. Not a stored passage, just a pointer used to grade whether the retriever fetched the right rules.

In the vector DB? **No**

So nothing "holds" the vector database except the rules. The label and the gold key are kept off to the side and only brought in when it is time to check the answer. That separation is the whole reason the study can claim a metric missed a legal error: the correctness answer never went near the retrieval the metric was scoring.